NETFLIX

Content Strategy & Data Analytics Report

Enterprise-Level Analysis | Python | Power BI | 8,807 Records

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At a Glance

8,807
Total Records

12
Features

40+
Years of Data

17
Visualizations

600%+
Content growth 2015–2020

>60%
Drama, Comedy & Docs share

US & India
Top 2 content markets

TV-MA & TV-14
Dominant audience ratings

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01 Business Problem Statement

Primary Research Question

What content investment patterns and strategic decisions can be inferred from Netflix's catalog data, and how can these insights guide future content strategy and market expansion?

Business Challenges Identified

- How to balance Movie vs. TV Show investment to maximize user retention and acquisition?
- Which geographic markets represent the highest growth opportunity for localized content?
- What content ratings and genre mix best serves Netflix's target demographic segments?
- At what rate is Netflix's content library growing, and is the growth sustainable?
- How should content duration and maturity ratings evolve to match audience preferences?

Scope & Limitations

Analysis based on content metadata only. The following data dimensions are not available:

- User engagement data (watch time, completion rates, repeat views)
- Revenue or subscription impact per title
- User review scores or sentiment data
- Production cost or licensing fee data

These gaps represent opportunities for future analysis. Engagement telemetry integrated with catalog metadata would enable predictive content investment models and significantly improve ROI per title.

02 Dataset Overview & Architecture

Dataset Structure

8,807	12	4,303	0
Total Records	Total Features	Missing Values	Duplicate Records

Column Schema

Column	Type	Description	Missing
show_id	Object	Unique identifier per title	0
type	Object	Movie or TV Show	0
title	Object	Name of the content	0
director	Object	Director name(s)	2,634
cast	Object	Main cast members	825
country	Object	Country of production	831
date_added	DateTime	Date added to Netflix	10
release_year	Int64	Original release year	0
rating	Object	Audience rating (TV-MA, TV-14, etc.)	4
duration	Object	Runtime (min) or seasons	3
listed_in	Object	Genre categories	0
description	Object	Short synopsis	0

03 Data Engineering & Cleaning Pipeline

Cleaning Pipeline Overview

Step 1

Initial Loading & Exploration

Shape check, info(), describe(), null counts, duplicate check → 0 duplicates found

Step 2

Missing Value Treatment

director, cast, country imputed with "Unknown". date_added filled using mode. Strategy: preserve all 8,807 rows

Step 3

Rating & Duration Fix

3 records had duration values misplaced in rating column. Corrected manually via show_id lookup and external verification

Step 4

Genre Normalization

listed_in split on comma; first genre retained as primary. year_added and month_added extracted for time series analysis

Data Engineering Outcome

✅ After cleaning: 0 duplicates, 0 nulls in all analytical columns except 4 in rating. Dataset fully ready for exploratory and advanced analysis.

Python Libraries Used

pandas

Data manipulation

numpy

Numerical operations

matplotlib

Static visualizations

seaborn

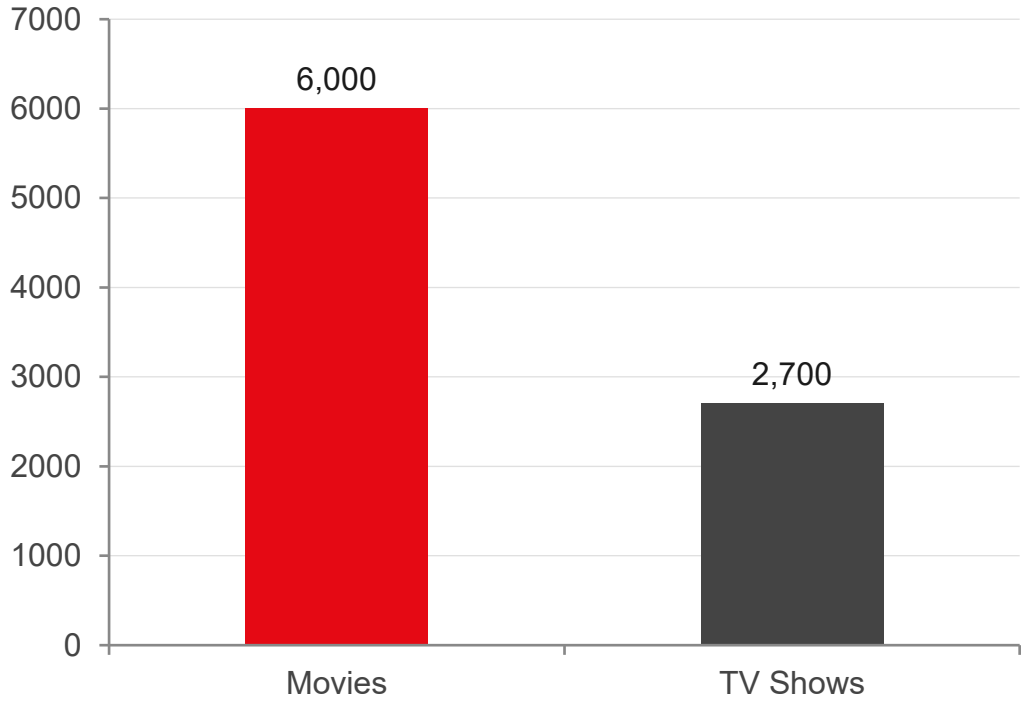
Statistical visualizations

plotly.express

Interactive visualizations

04 Exploratory Data Analysis

4.1 Content Type Distribution



~6,000

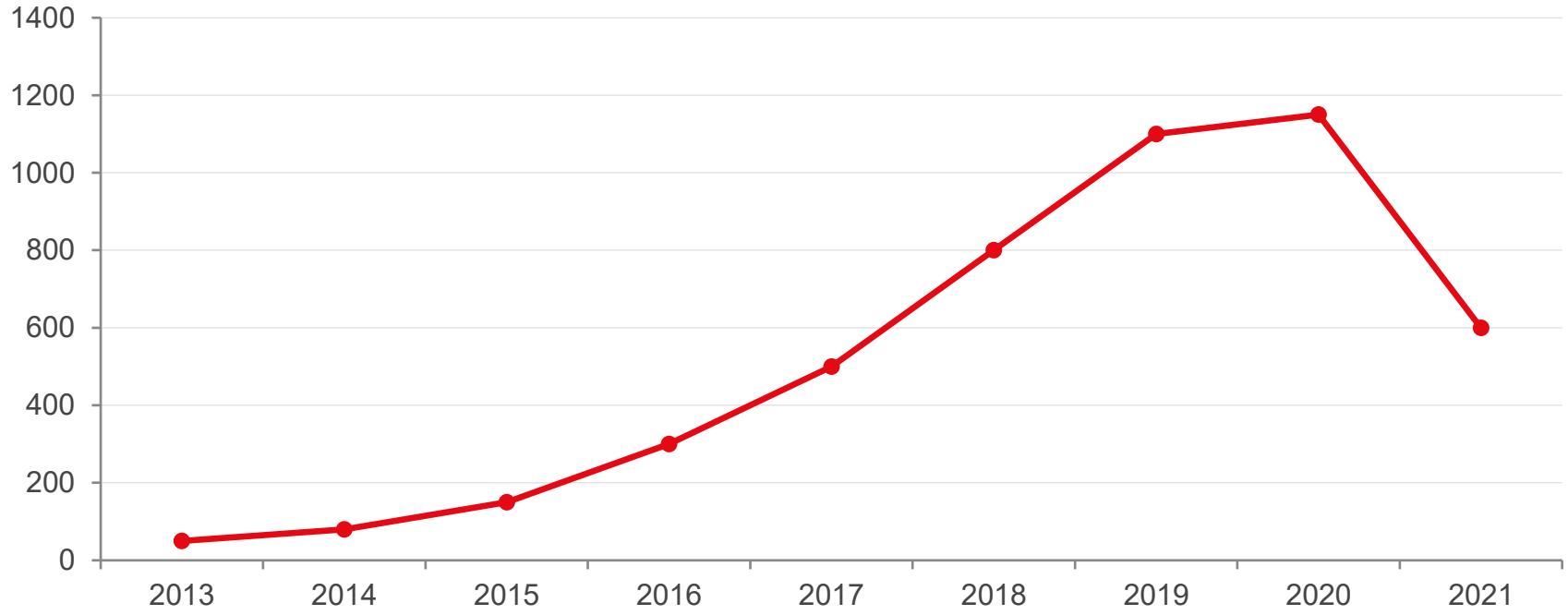
Movies (68%)

~2,700

TV Shows (32%)

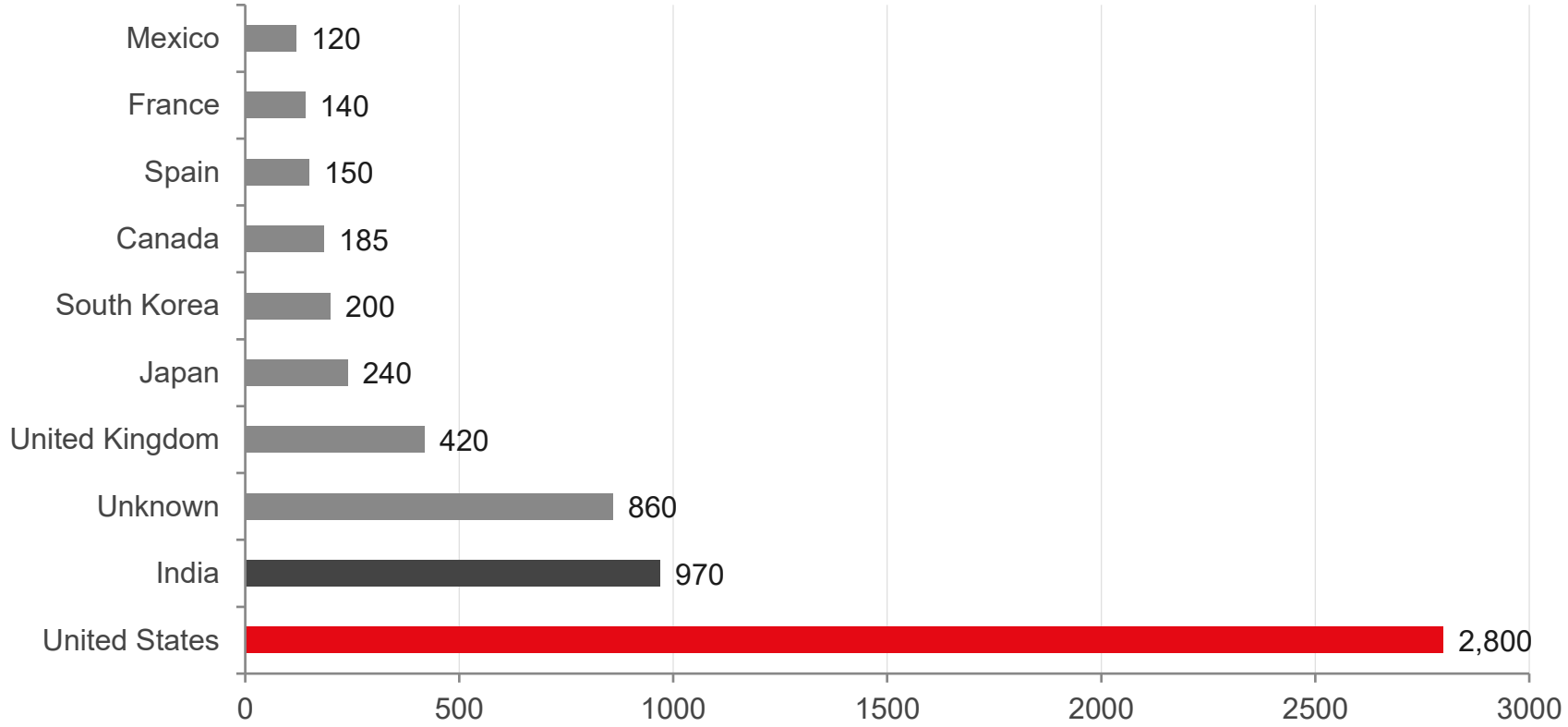
💡 **Insight:** Movies serve as the acquisition engine. TV Shows serve as retention tools. A healthy split strategy requires both.

4.2 Content Release Year Trend



💡 **Insight:** Sharp concentration from 2013 onwards, peaking 2019–2020. Reflects aggressive original content production ramp-up. Post-2020 decline signals strategic stabilization.

4.3 Geographic Content Distribution — Top 10 Countries



4.3 Geographic Insights

💡 **Insight:** India's strong #2 position reflects Netflix's investment in Bollywood and regional content as part of its emerging market strategy. Japan and South Korea signal the K-Drama and Anime opportunity.

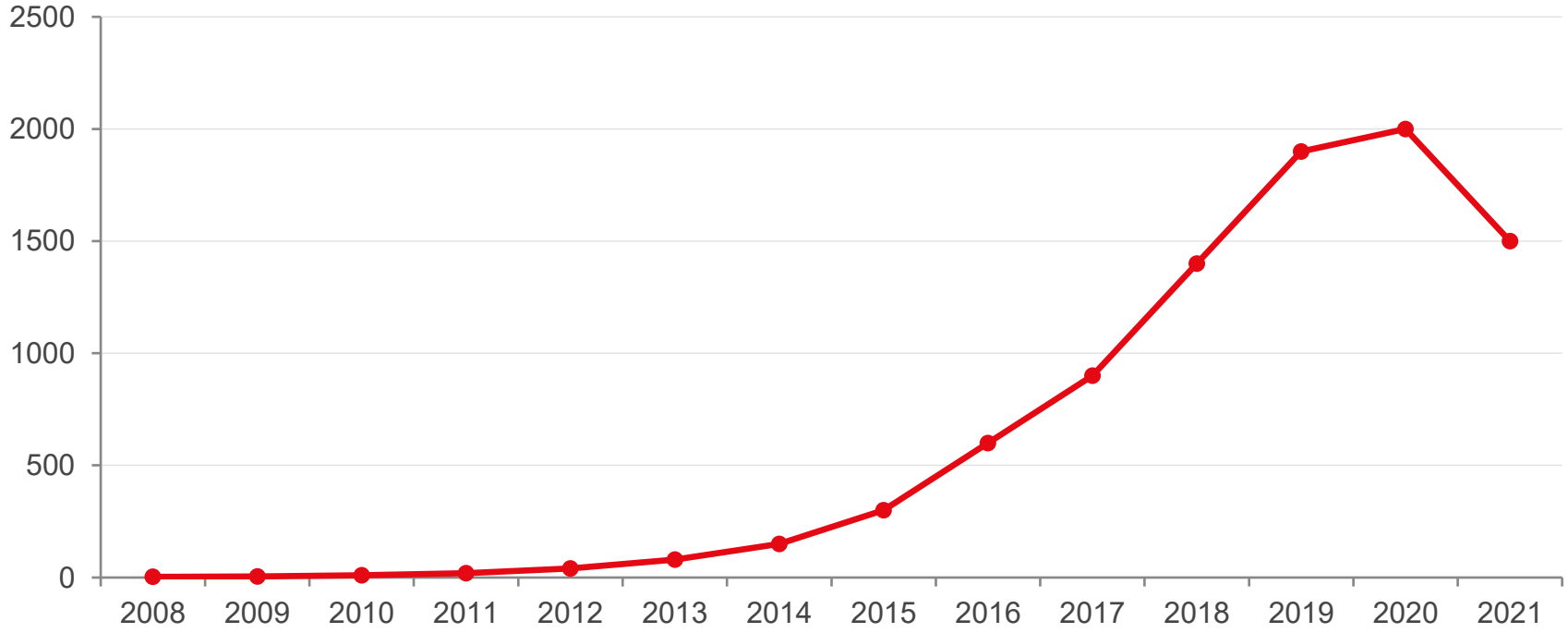
- United States: ~2,800 titles — dominant content producer, nearly 3× the second country
- India: ~970 titles — fastest-growing non-English market, Bollywood + regional originals
- "Unknown" (~860): represents titles without country metadata — a data quality gap
- United Kingdom: ~420 titles — premium drama and comedy co-productions
- Japan (~240) & South Korea (~200): K-Drama and Anime-adjacent expansion opportunity
- LATAM (Mexico) and Europe (Spain, France) show emerging production capacity

05 Time Series Analysis — Content Growth

5.1 Four Strategic Growth Phases

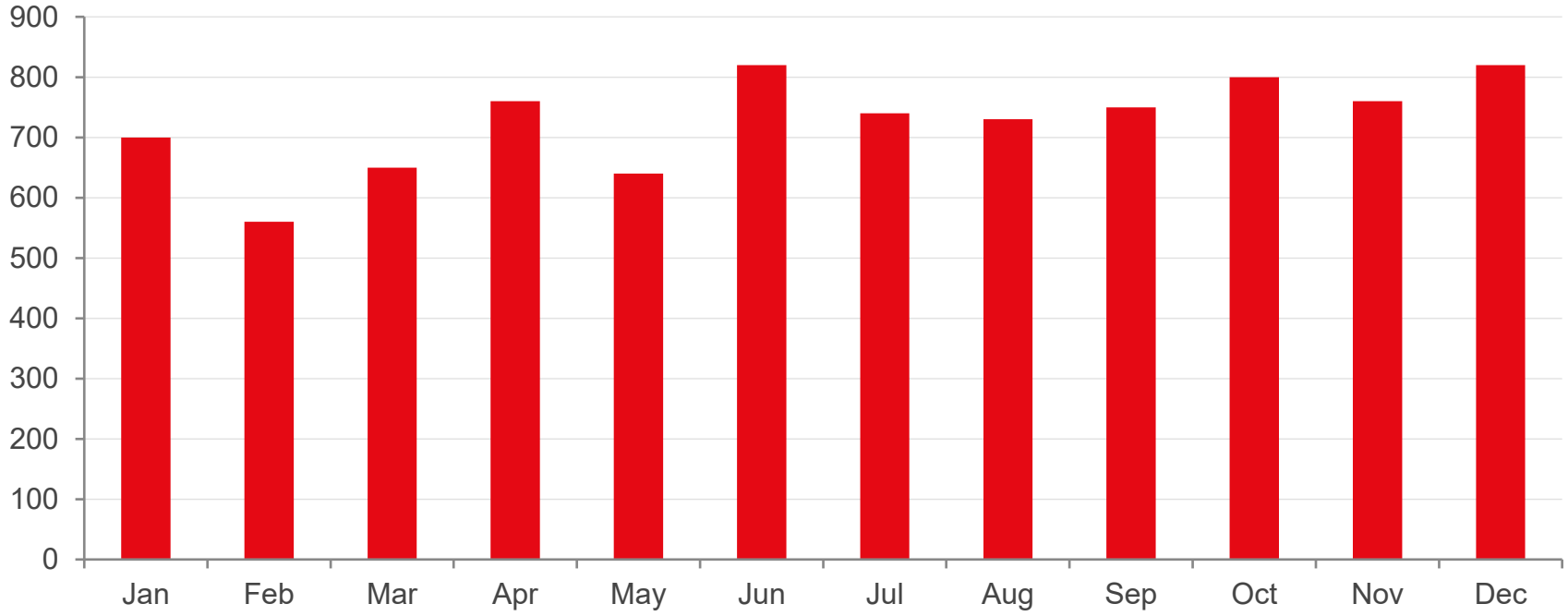
Phase	Period	YoY Growth	Strategic Context
Foundation	2008–2014	Slow & Steady	Initial streaming library build-up
Acceleration	2015–2017	+50–80%	International expansion begins
Peak Expansion	2018–2020	+100–150%	Original content investment surge
Stabilization	2021+	Declining	Shift from volume to quality curation

5.1 Year-Wise Content Addition Trend



💡 **Insight:** YoY growth peaked at 150%+ in 2019–2020. Post-2021 stabilization indicates a strategic pivot from aggressive expansion to content quality optimization.

5.2 Monthly Content Addition Pattern



💡 **Insight:** Content additions spike in Q4 (October–December), aligning with holiday season viewing. This is a deliberate programming strategy to maximize subscriber engagement during peak viewing months.

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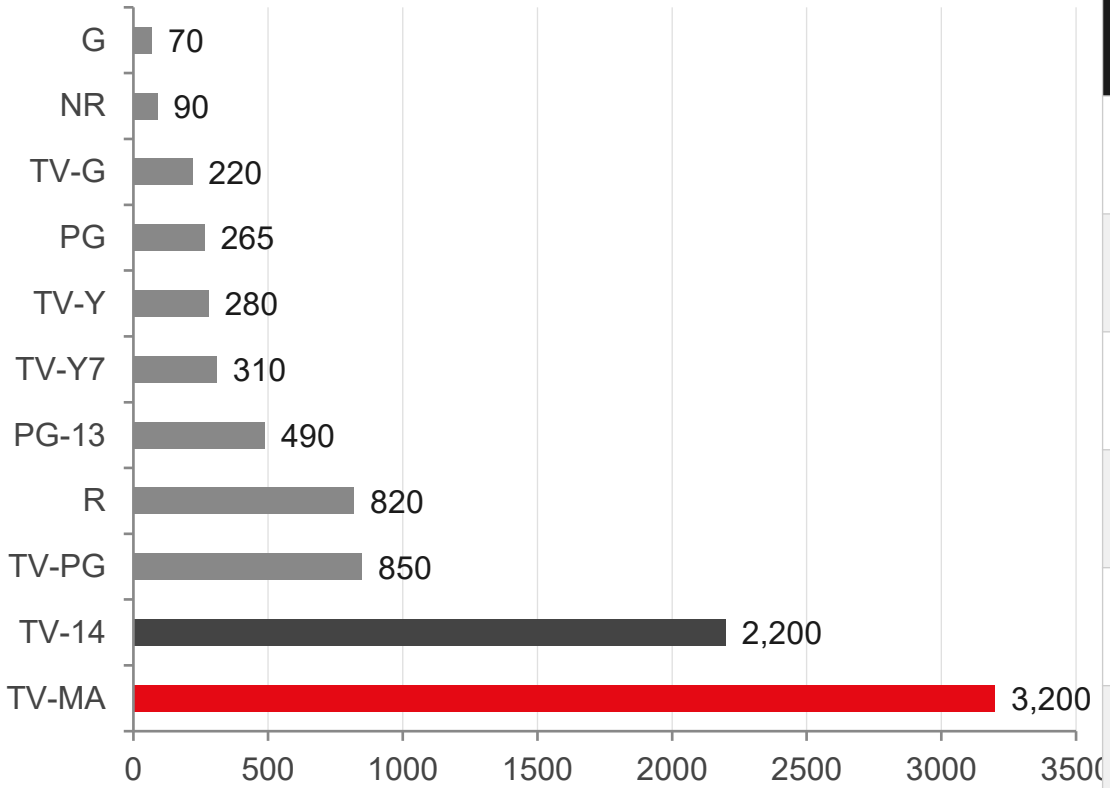
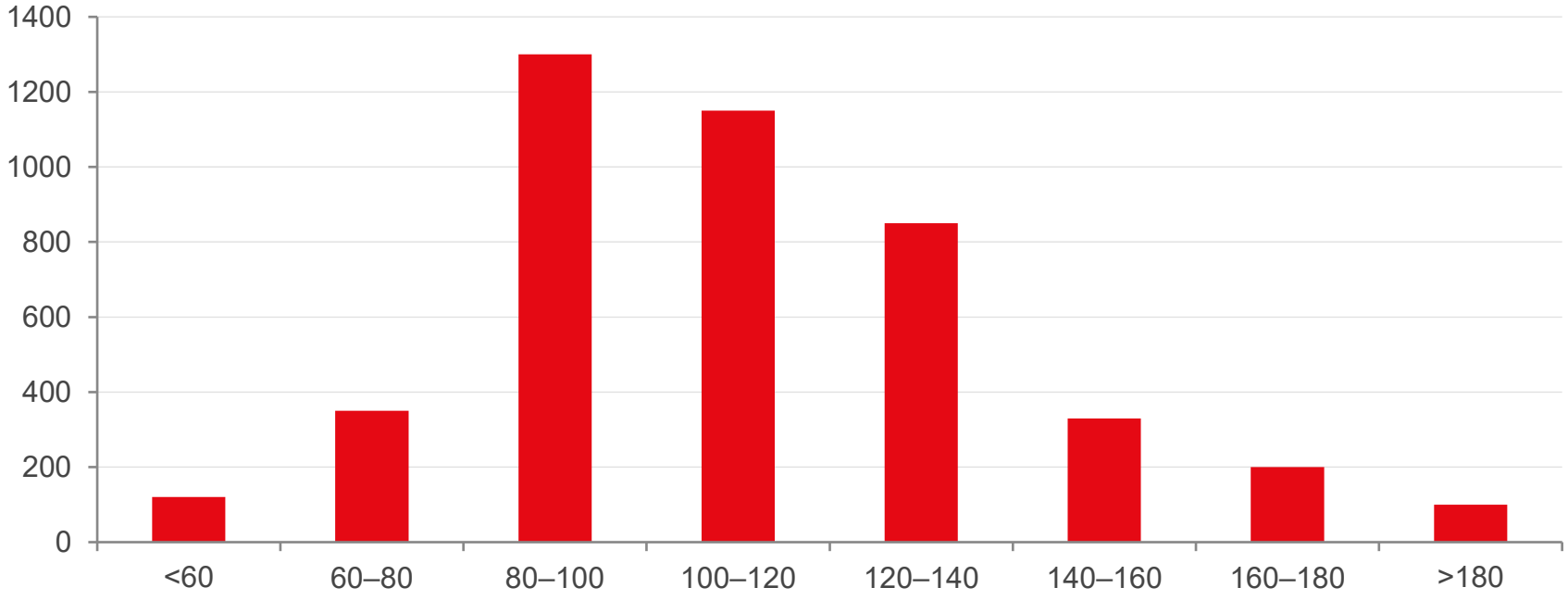
Content Analysis — Ratings & Duration

6.1 Content Rating Distribution



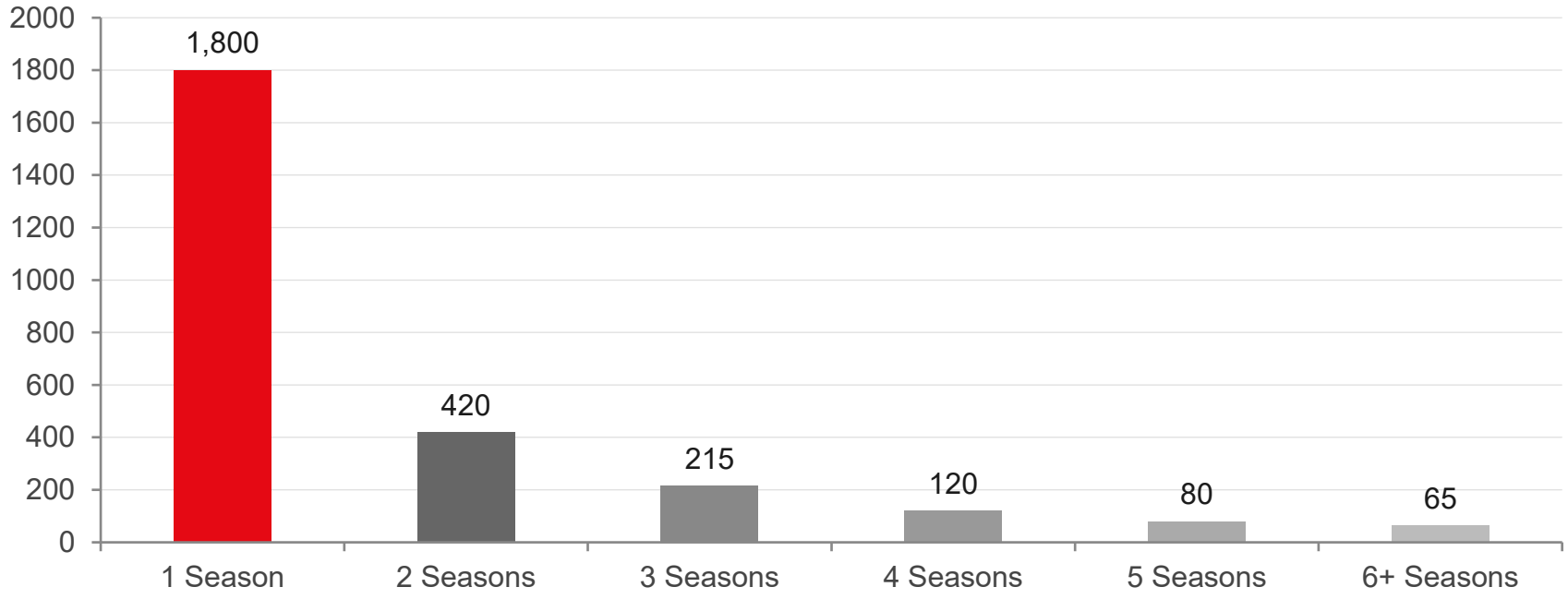
Rating	Audience	Count	% Library
TV-MA	17+	3,200	36.3%
TV-14	14+	2,200	25.0%
TV-PG	Parental Guidance	850	9.7%
R	17+ (film)	820	9.3%
PG-13	13+	490	5.6%
Others	Various	1,247	14.1%

6.2 Movie Duration Distribution



💡 **Insight:** Most Netflix movies fall in the 80–120 minute range — the widely accepted "completion sweet spot". This reflects deliberate content acquisition criteria targeting high completion rates.

6.3 TV Show Season Distribution

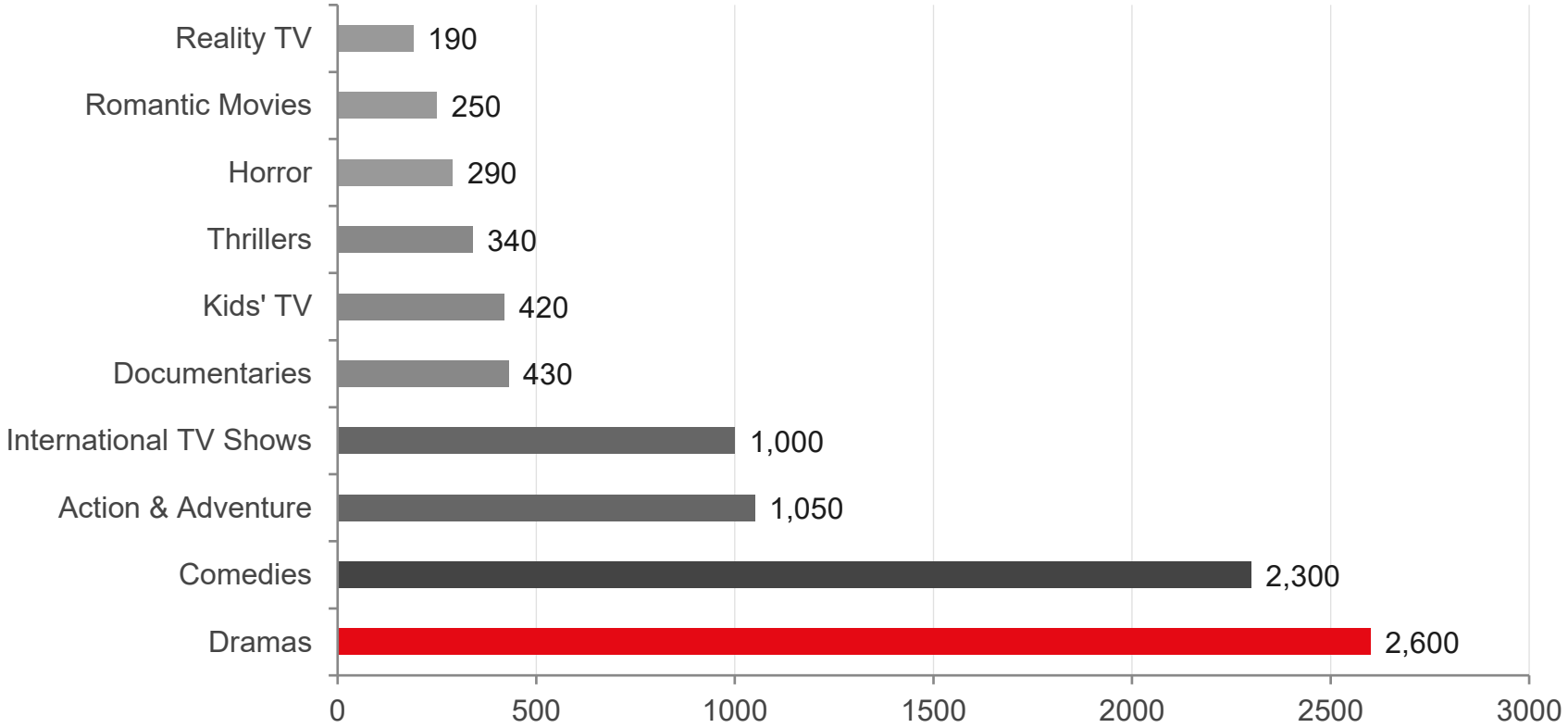


💡 **Insight:** Single-season TV shows dominate the library. Netflix uses limited series as a low-risk format. Multi-season renewals are reserved for proven performers — a data-driven investment approach.

07

Genre Analysis & Trends

7.1 Top Genre Distribution

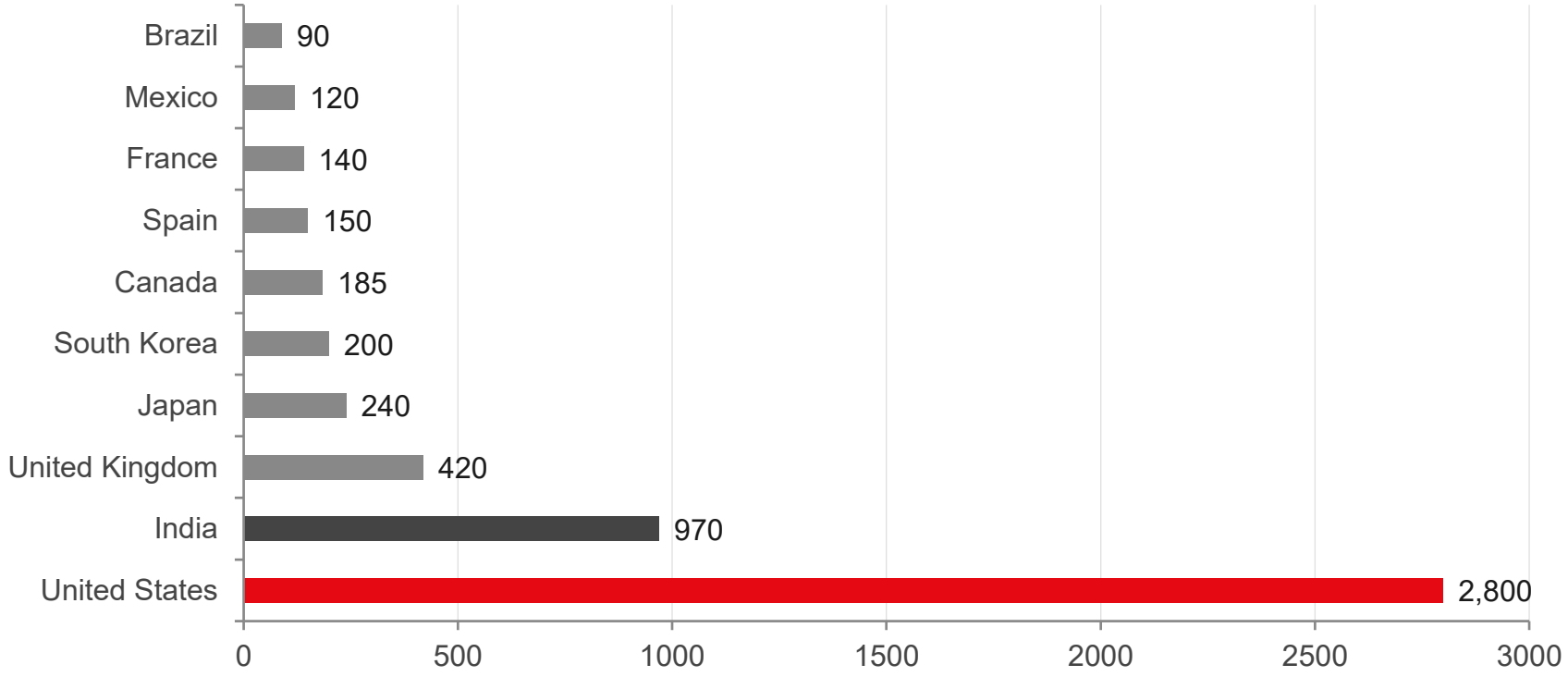


7.2 Genre Trend Analysis & Business Implications

Genre	Trend Direction	Business Implication
Dramas	Consistently High ↑	Core content pillar — stable investment zone
International TV Shows	Fastest Growth 🚀	Key indicator of global expansion success
Comedies	Steady ↑	Mass appeal genre — broad demographic reach
Documentaries	Slow but Consistent ↑	Appeals to educated, high-engagement audience
Action & Adventure	Volatile →	High cost, high reward — selective investment

08 Geographical Analysis & Market Intelligence

8.1 Regional Content Concentration



💡 **Insight:** US leads at ~2,800 titles — nearly 3× India. The gap is narrowing as emerging markets gain investment.

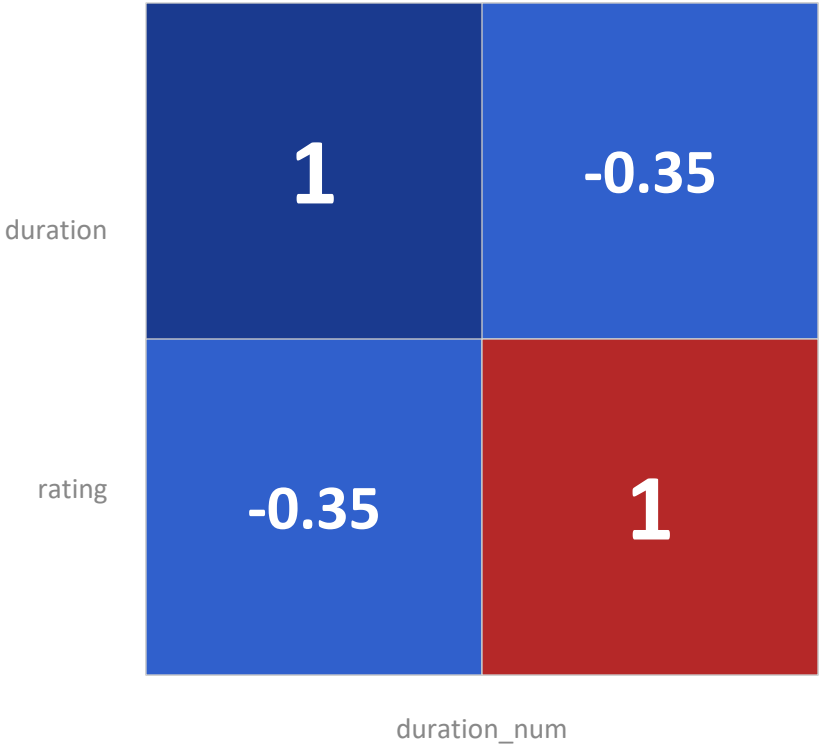
8.2 Emerging Markets Analysis

💡 **Insight:** India, Japan, South Korea, and Mexico are Netflix's highest-priority regional content markets. Regional content achieves global audience appeal, offering outsized ROI on production investment.

- India: Fastest compound growth rate — Hindi, Tamil, and Telugu originals driving APAC subscriber acquisition
- South Korea: K-Dramas achieve global crossover appeal (Squid Game: most-watched series in history)
- Japan: Anime-adjacent content draws global niche audiences with very high engagement and loyalty
- Mexico & LATAM: Growing telenovela and thriller production capacity — LATAM market under-represented
- Spain & France: Co-production treaties enable cost-effective European content
- MENA region: Still significantly under-represented — high-growth opportunity market

09 Correlation & Advanced Statistical Analysis

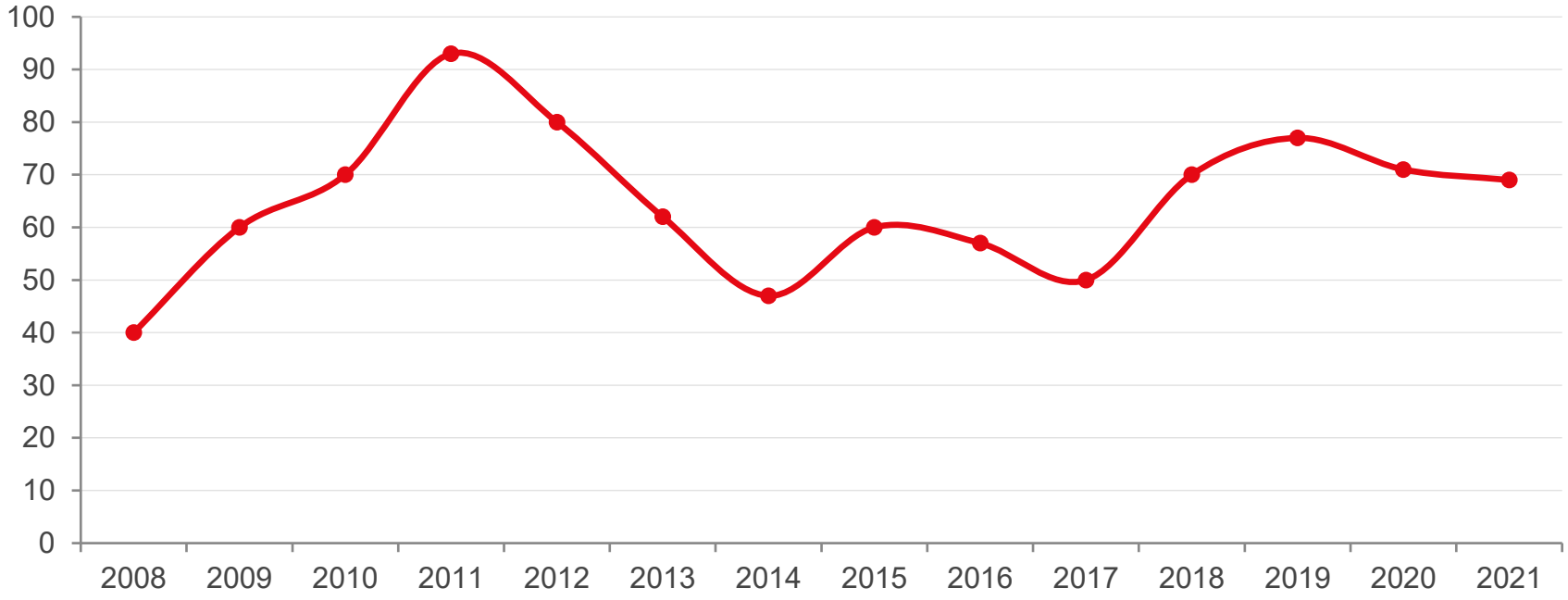
9.1 Duration vs. Rating Correlation



Key Finding

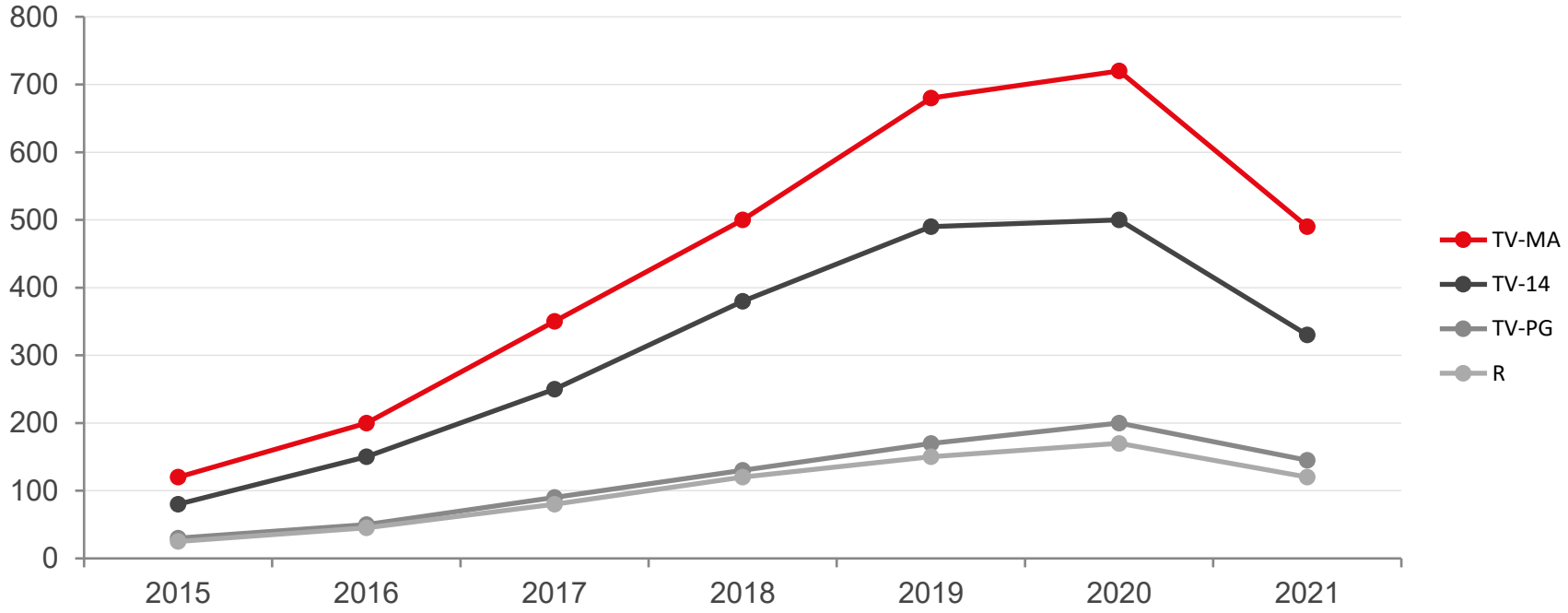
- Correlation coefficient: -0.35 (weak negative)
- Content length does NOT significantly dictate its audience rating category
- Short and long content alike spans all rating tiers
- Duration alone is not a predictor of content maturity classification
- Validates Netflix's approach: content for all lengths across all demographics

9.2 Average Content Duration Over Time



💡 **Insight:** Duration peaked ~2011 (early library built from longer films), dipped ~2014 (TV show expansion), stabilized ~70 min average. Recent stabilization reflects mix of shorter originals and feature-length films.

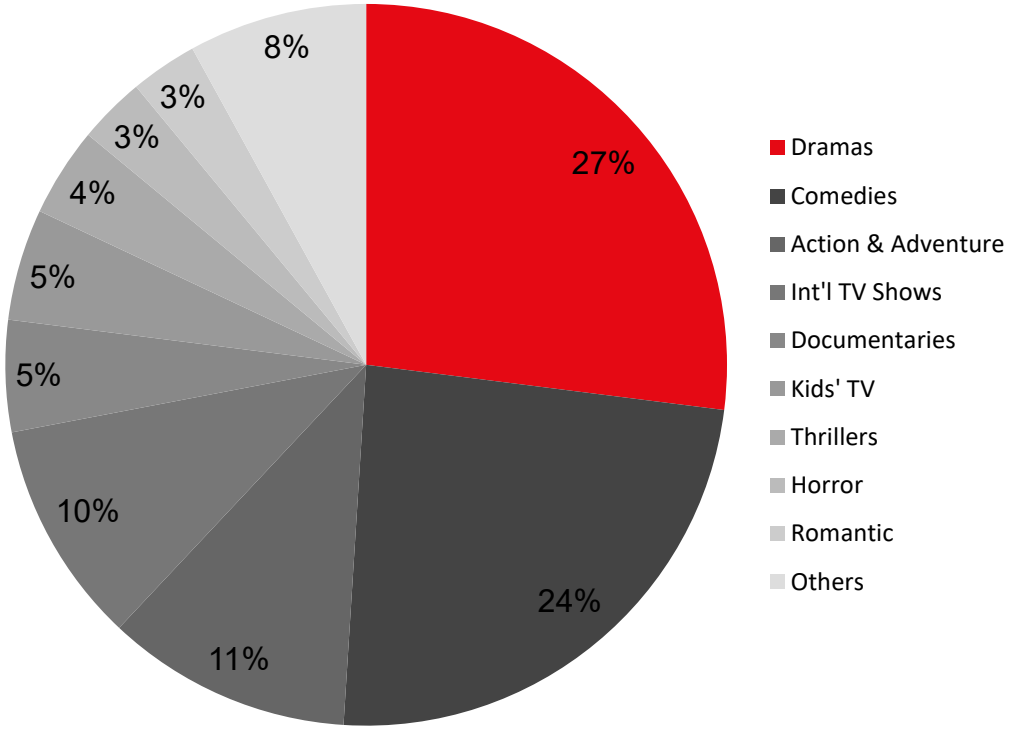
9.3 Rating Evolution Over Time



💡 **Insight:** TV-MA shows steepest growth — Netflix deliberately targets mature adults with higher engagement, longer sessions, and lower churn. Aligns with binge-watching content strategy.

10 Content Variety & Audience Engagement

10.1 Genre Diversity Distribution



- Drama & Comedy = 51% of total library
- Drama, Comedy & Docs = >60% combined
- International TV Shows: highest growth
- Children's content only ~5% — underserved
- Horror & Thriller: niche but loyal audiences

10.2 User Preference Inference

Without direct engagement data, user preference is inferred from content availability patterns.

Movies

= Acquisition Engine

- Broad library appeal
- New subscriber attraction
- ~6,000 titles (68%)

TV Shows

= Retention Tool

- Multi-episode engagement
- Subscription stickiness
- ~2,700 titles (32%)

💡 **Insight:** Netflix must maintain both content types to simultaneously attract new subscribers and retain existing ones. A healthy split strategy is critical for long-term growth.

11 Power BI Dashboard Overview

11.1 Dashboard Components

Dashboard Panel	Chart Type	Key Metric	Business Use
Content Growth Timeline	Line Chart	Titles added per year	Track expansion phases
Genre Breakdown	Donut Chart	% share by genre	Portfolio balance
Geographic Heatmap	Filled Map	Titles by country	Market prioritization
Rating Distribution	Stacked Bar	Audience rating split	Demographic targeting
Movie vs TV Trend	Area Chart	Type split over time	Investment mix KPI
Duration Analysis	Histogram	Runtime distribution	Content format strategy
Top Directors	Bar Chart	Content volume by director	Talent relationship mgmt
Monthly Add Cadence	Column Chart	Monthly addition pattern	Content scheduling

11.2 Power BI Key Skills Demonstrated

- Data import and transformation using Power Query (M language)
- DAX calculated columns and measures for YoY growth, % share, rolling averages
- Interactive slicers for Type, Country, Rating, Year, and Genre
- Cross-filtering between visuals for multi-dimensional drill-down
- Custom color theming aligned with Netflix brand palette (#E50914)
- Bookmarks and navigation for multi-page dashboard flow

File: Netflix_title_dashboard.pbix | Requires Microsoft Power BI Desktop | Dataset: netflix_titles.csv

12 Strategic Recommendations

12.1–12.3 Recommendations (Part 1)

12.1 — Regional Content Investment

Accelerate Emerging Market Production

India shows fastest YoY growth. Doubling investment in Hindi, Tamil, and Telugu originals — plus K-Drama and Anime partnerships — projected to unlock 50–100M new subscribers in APAC and LATAM.

12.2 — TV Show Retention Strategy

Increase Multi-Season Series Investment

Most TV shows are single-season. A 2-season minimum commissioning strategy for top-performing originals would materially reduce subscriber churn and increase platform stickiness.

12.3 — Underserved Segments

Expand Children's & Family Content

Only ~5–8% of library carries G, TV-Y, or TV-Y7 ratings. Family households have high retention rates. Animated originals would reduce churn and increase household account value.

12.4–12.6 Recommendations (Part 2)

✓

12.4 — Genre Core Optimization

Protect and Deepen the Drama/Comedy Core

Drama and Comedy represent >60% of catalog. Data supports a "core-and-explore" strategy: protect the drama/comedy core while using 10–15% of budget to test high-upside categories (Documentary Series, Interactive Content).

✓

12.5 — Mature Content Balance

Rebalance Rating Skew with PG-13 and TV-PG

Over 61% of content targets TV-14 and TV-MA. A targeted 15% shift toward PG-13/TV-PG in new commissioning could broaden the serviceable audience without alienating the core adult demographic.

✓

12.6 — Data Capability Enhancement

Integrate Engagement Metrics into Content Analytics

Current dataset lacks user behavior signals. A unified content analytics platform combining catalog metadata with engagement telemetry would enable predictive investment models and significantly improve ROI per title.

13 Conclusions

What the Data Confirms & Opportunities Identified

✓ What the Data Confirms

- Content library grew 600%+ post-2015
- Drama and Comedy are resilient, scalable genres
- India is fastest-growing content market
- Movies used for acquisition, shows for retention
- TV-MA dominance confirms adult-first strategy

⚠ Opportunities Identified

- Underserved children's & family segment
- Over-reliance on single-season TV shows
- No engagement data for true ROI measurement
- Rating skew toward mature content only
- LATAM and MENA markets under-represented

Strategic Conclusion

"Netflix built its position through volume and consistency, but sustaining that position at scale requires a pivot toward precision — investing in the right content, in the right markets, for the right audiences, with engagement data at the core of every decision."

💡 **Insight:** This analysis reveals that Netflix's content strategy is not opportunistic — it is highly deliberate and data-informed. The platform has methodically scaled from a US-centric service to a global content powerhouse by following a clear strategic playbook.

Appendix — Technical Reference

A. Python Libraries

pandas	— Data manipulation
numpy	— Numerical operations
matplotlib.pyplot	— Static visualizations
seaborn	— Statistical visualizations
plotly.express	— Interactive visualizations

B. Data & Tools

- Dataset: Netflix Titles Dataset (Kaggle) | 8,807 records | 1925–2021
- Python Version: 3.9+ | pandas ≥ 1.3, matplotlib ≥ 3.4, seaborn ≥ 0.11, plotly ≥ 5.0
- Power BI: Netflix_title_dashboard.pbix — Power BI Desktop required (free from Microsoft)

Thank You

Shivam Yadav | Data Analyst

Python · Power BI · Pandas · Matplotlib · Seaborn · Plotly